

Splash StableSwap (ADA/OADA) exploit: incident report and funds trace

Splash team, 13 September 2026, 16:55 UTC.

Sources: Cardano mainnet (public chain data), the `splash-core` repository, the mainnet deployment manifest, the Anastasia Labs audit, and the public exchange-address datasets named in section 6.4. We rebuilt the deployed script from source and matched it byte for byte against the chain. All addresses and transactions link to Cexplorer.

1. TL;DR

- On 13 Sep 2026, between 00:47 and 00:48 UTC, one actor drained the Splash `stableFnPoolT2t` ADA/OADA pool in two transactions. The second one took 2,434,648.42 ADA and 1,988,222.18 OADA out of the pool (2,424,778.42 ADA net of the attacker's own 9,870 ADA deposit). The LP supply did not change.
- The cause is in the validator. We rebuilt the deployed bytecode from source (commit `0d28fe4`, Aiken 1.0.24; the swap branch is identical to the audited code). The validator subtracts accrued protocol fees from the real balances to get a "tradable reserve" and never checks that the result is positive. The attacker first inflated the fee counter, which the code allows because the fee check is only a lower bound and `protocol_fee_num = 0`, then spent the real ADA balance below the counter. The invariant polynomial accepted a negative reserve and let both assets leave.
- 2,552,176.54 ADA went to six deposit addresses (five fresh, one used once before on 9 Sep) at four custodial clusters and was swept into their consolidation wallets within 82 minutes of transaction B. Two clusters are attributed to exchanges by two independent public datasets each plus direct chain links (section 6.4): KuCoin, 786,851.54 ADA, and Gate.io, 550,000 ADA. The other two clusters, 1,215,325 ADA, are custodial systems whose operator we have not confirmed. Cluster C's 550,000 ADA has moved on to a large custodial wallet; cluster B's 665,325 ADA is still sitting in its consolidation wallet. All the OADA was dumped into a thin OADA/FLDT pool on Minswap and converted to ADA, and the attacker got only about 115k ADA for it. OADA has no other exit. As of 16:20 UTC 1.76M OADA sits in that Minswap pool at a small fraction of its peg, and any new ADA/OADA liquidity would be arbitrated against it (section 7).
- The attacker's seed money, 12,186 ADA, was withdrawn from a KuCoin withdrawal wallet 17 minutes before the exploit, and 786,851 ADA went back to KuCoin. The KuCoin deposit address that received 236,851 ADA had been used once before, on 9 Sep 2026. Exchanges assign deposit addresses per account, so we are pursuing this with the exchange (section 6.5).

2. Affected component

Item	Value
Validator	<code>stableFnPoolT2t</code> (Splash StableSwap, two-asset, "exact" variant)
Script hash	<code>5d3df99cfbbf282bd76a3d76a2e30bdd22e61c56f1462447938933b</code>
Reference script UTxO	<code>43da5e009b4c7d8594c5dc51d3901b89da9d64d208b7b2fa8980a90c42601132#0</code>
Pool address	<code>addr1x9wnm7vle7al9q4aw63aw63wxz7aytnpc4h3gcjy0yufxwaj764lvrxdayh2ux30fl0ktuh27csgmpevdu89jlxppvr s84l0h4</code>
Pool NFT	<code>66eabfc9d78f7bbfb4a5ad9ae7de59f0a8bebcc96f0f57c33044f110.6e6674</code>
LP token	<code>153c2bfbe184b1edc7a0d89bca1608b1dcc24535102fa1f732a9ca2.6c71</code>
OADA	<code>f6099832f9563e4cf59602b3351c3c5a8a7dda2d44575ef69b82cf8d</code> . (empty name, 6 decimals)

Script hash and reference UTxO match `bloom-cardano-agent/resources/mainnet.deployment.json` (`stableFnPoolT2t`) at commit [f2312642](#) of `splash-offchain-multiplatform`. The manifest lists only this one stable-pool validator (plus its deposit/redeem order scripts).

Which source is deployed. We reproduced the on-chain script byte for byte. The script (Plutus V2, 3,177 bytes, reference UTxO created 28 Jun 2024 15:27 UTC) is the build of `validators/stable_pools/t2t/pool_t2t_exact.ak` at `splash-core` commit [0d28fe4](#) ("Softer condition for LP validation", 28 Jun 2024 06:33 UTC, the last stable-pool commit before deployment), compiled with Aiken v1.0.24-alpha+982eff4 (stdlib 1.7.0), with the validator parameter `dao_voting_witness` applied as 28 zero bytes (`581c00...00`, read back from the outermost application in the decoded on-chain UPLC). `aiken build + aiken blueprint` apply on that commit yields hash [5d3df99fcfbf282bd76a3d76a2e30bdd22e61c56f1462447938933b](#) and a `compiledCode` identical to the on-chain bytes. Neighbouring commits do not match (`64ed87e` gives `ac309ef2...`; the pre-fix code at `b645822` gives `e08171b2...`, 3,118 bytes), and Aiken 1.0.26 produces a different artefact (2,956 bytes), so the identification is unique. Between the squash-merge of the stable-pool work ([8acf906](#), 26 Jun, into the `stable-swap-audit` branch; file identical to branch commit [b645822](#)) and the deployed [0d28fe4](#) the validator differs in one hunk in the deposit/redeem branch (LP-delta rounding tolerance, introduced in [64ed87e](#) and relaxed in [0d28fe4](#)) plus added tests. The swap branch executed by transactions A and B is unchanged. The `validators/stable_pools/t2t/` sources live on the `stable-swap-audit` branch; the newer `validators_v2/validators/stable_pool/pool.ak` on `main` is not what the pool ran.

3. Why it was possible: the exploit logic

The validator computes tradable reserves as

```
x_tradable = x_total - protocol_fees_x
y_tradable = y_total - protocol_fees_y
```

and then runs the StableSwap invariant checks on them. Three checks are missing, and together they open the hole:

1. There is no domain check. Nothing requires `x_tradable > 0` and `y_tradable > 0` (before or after the action), nor `protocol_fees < total`. An intermediate negative integer is perfectly legal in Plutus even though the ledger never holds negative balances.
2. Fee accrual is bounded only from below. In `pool_t2t_exact.ak`:

```
valid_quote_protocol_fees = quote_protocol_fees_delta * f_rev >= -quote_total_delta * protocol_fee_num
```

With the pool's `protocol_fee_num = 0` the right-hand side is 0 (`f_rev = 100,000 - lp_fee_num = 99,900`), so any non-negative fee delta passes. The attacker can declare almost the whole deposit as "protocol fees".

3. There is no direction check. A "swap" is chosen by the sign of `y_tradable_delta`; there is no requirement that the base asset actually enters the pool and only the quote leaves. A transition in which both tradable reserves decrease is not rejected.

The invariant polynomial (with `an2n = 3200`, unit multipliers) is

```
F(D, x, y) = D³ + 3196·x·y·D - 3200·x·y·(x + y)
```

checked as `F_before ≥ 0`, `F_after ≤ 0`, `F_after_shifted ≥ 0` (in the source `F_after` uses the quote reserve net of the LP-fee term, `y' = y_after - [Δy·lp_fee_num / (lp_fee_num - 100,000)] - 1`, and `F_after_shifted` uses `x·(y' - 2)` and `y' - 2`). These sign tests assume `x, y > 0`. With `x < 0` the products change sign and the checks are satisfiable for an economically absurd transition where the pool loses both assets. Passing the polynomial test therefore no longer guarantees that liquidity is preserved.

Step A: preparation (tx [bfb002ac...7f5bae2](#))

13 Sep 2026 00:47:36 UTC, block 13933439, AMM/swap branch, redeemer `PoolRedeemer(pool_in_ix=1, pool_out_ix=0, AMMAction([D=5854566784562]))`.

- The pool receives 9,870 ADA and 0.000001 OADA.
- Datum `protocol_fees_x` goes from 0 to 9,869,999,999 lovelace. The tradable ADA reserve grows by exactly 1 lovelace; everything else is booked as fees.
- LP balance unchanged (9223366214166485960).

Step B: withdrawal (tx [47dae2da...7682d0](#))

00:48:21 UTC, block 13933444 (45 s later), same branch, redeemer `PoolRedeemer(pool_in_ix=0, pool_out_ix=0, AMMAction([D=5854566784564]))`, spends A#0 (pool) and A#1 (attacker).

Pool state	Before A	After A	After B
Real ADA	2,424,788.424937	2,434,658.424937	10
Real OADA	3,430,000.064178	3,430,000.064179	1,441,777.879782
<code>protocol_fees_x</code> (ADA)	0	9,869.999999	9,869.999999
Tradable ADA	2,424,788.424937	2,424,788.424938	-9,859.999999
LP tokens in pool	9223366214166485960	same	same

Both transactions executed with `valid_contract: true`. Pool address, NFT, LP balance and all other datum fields are preserved. Nothing was minted or burned.

Transaction B took 2,434,648.424937 ADA and 1,988,222.184397 OADA out of the pool. After their own 9,870 ADA deposit and 1.35 ADA of network fees the attacker netted 2,424,777.072417 ADA and 1,988,222.184396 OADA.

The pool is left with a negative tradable ADA reserve and cannot operate normally.

Verification

- Both transactions re-fetched from the chain; all balances, datums and redeemers above match.
- Deployed bytecode reproduced from source (see section 2): commit [0d28fe4](#), Aiken 1.0.24-alpha, parameter `00x28`; hash and bytes identical to the script at [43da5e00...#0](#).
- Independent numeric replay of the swap branch of the deployed source on the on-chain values (Python big integers, floor division as in Plutus `divideInteger`; base/quote chosen by the sign of `y_tradable_delta` as in the code):
 - Tx A: base = OADA, quote = ADA. The fee check `valid_quote_protocol_fees = 9,869,999,999.99,900 ≥ -9,870,000,000.0` holds, `non_quote_protocol_fees_preserved` holds, and the three invariant checks `F_before ≥ 0`, `F_after ≤ 0`, `F_after_shifted ≥ 0` hold with `D = 5,854,566,784,562`. Accepted.
 - Tx B: base = ADA, quote = OADA. Tradable ADA goes from 2,424,788,424,938 to -9,859,999,999. The fee check `0 ≥ 0` holds, the non-quote check holds, and the three polynomial checks hold with `D = 5,854,566,784,564`. Accepted.
 - With a domain check (`tradable > 0` before and after, `fees < total`) B is rejected; with a two-sided fee bound on the fee delta A is rejected. Either fix alone stops the exploit.
- Full pool history from block 13,900,000 (5 Sep 2026): the six transactions before A (6–11 Sep) are routine deposits, redeems and one swap through the normal batcher; `protocol_fees_x` was 0 throughout. There were no test transactions against this pool on mainnet since 5 Sep. The attacker's only mainnet preparation was acquiring 100 OADA (tx [2f75614c...](#), 100 ADA into the Optim OADA contract) 26 s before A.

4. Audit

- Anastasia Labs, "Security Audit Report, Splash Protocol Stableswap", v1.0, dated 5 August 2024. Engagement 7 June – 19 July 2024. Repository `splashprotocol/splash-core`, audited commit `d9768bb4c5b775c2ad9d7e2755eb8a5d743dd25f`. Five files audited: `validators/stable_pools/t2t/pool_t2t_exact.ak` (the deployed logic), `pool_t2t.ak`, `proxy_dao_t2t.ak` and the two three-asset variants; the deposit/redeem order scripts were out of scope.
- Listed on <https://docs.splash.trade/concepts/audits>; PDF: [anastasiabooks_stableswap.pdf](#)
- The SHA-256 the audit lists for `pool_t2t_exact.ak` (`e6a0596f...661fa`) matches the file at the audited commit `d9768bb4` exactly. Between the audited commit and the deployed commit `0d28fe4` the validator changed in two places only: a test-template constant (`8acf906`) and the LP-delta tolerance in the deposit/redeem branch (`64ed87e`, relaxed in `0d28fe4`). The swap branch that executed transactions A and B was within the audited scope and is unchanged since the audited commit.
- Eight findings (ID-501 critical: DAO action validator misuse; ID-401 major: unrestricted DAO change; ID-301/302 medium: zero-spam, invariant at DAO change; ID-201/202 minor: dust attack, `lp_fee + protocol_fee > denom`; ID-101/102 informational), six marked resolved and two acknowledged. None of them concerns fee accrual in the swap branch, the sign of tradable reserves, or swap direction.

5. Attacker addresses

Role	Address
Attacker (exploit + all exits), stake key <code>stake1uyw7cns4wduxgkw2dymgchtpk4zxaxekc6aemutp44k8ffqhf2a3u</code>	<code>addr1q8a3zkth4xk7wvv4ehxa4w8mqtfp0mzhqxnrsd08tk0jl4qaa38p2umcv3vu56fk33wkrd2yd6dnd34mnhckrttvjjqtreakak</code>
Minswap order script address used by the attacker (same stake key)	<code>addr1z8p79rpkdz8x9d6tft0x0dx5mwuzac2sa4gm8cvkw5hcnqaa38p2umcv3vu56fk33wkrd2yd6dnd34mnhckrttvjjqxhlhxx</code>

The attacker address is fresh: first funded 00:30:05 UTC on 13 Sep, 13 transactions in total, current balance 1.544943 ADA. Everything has been moved out.

6. Full timeline and funds trace (all times UTC, 13 Sep 2026)

6.1 Seed and exploit

Time	Tx	What
00:30:05	<code>acc883fd...8230fa0</code>	12,186.35 ADA withdrawn from KuCoin withdrawal wallet <code>addr1qywgh46dqu7...</code> to the attacker (KuCoin attribution: section 6.4)
00:46:36	<code>ba5e1048...2370e61e</code>	split: 10 ADA collateral + change
00:47:10	<code>2f75614c...0b94bc9a</code>	100 ADA → 100 OADA via Optim contract <code>addr1z9ny26wa...</code>
00:47:36	<code>bfb002ac...7f5bae2</code>	Tx A: fee counter inflated (block 13933439)

Time	Tx	What
00:48:21	47dae2da...7682d0	Tx B: pool drained (block 13933444)

6.2 ADA exits (2,436,851.54 ADA + FLDT proceeds)

Each destination is a deposit address (five fresh; D5 had been used once before, on 9 Sep, section 6.5) that was swept within minutes into a custodial cluster. Cluster labels are established in section 6.4; only clusters A and D are attributed to named exchanges (KuCoin, Gate.io).

Time	Tx	Amount	Destination (deposit address)	Cluster
00:54:44	8ca45d40...9fa1af8d3	550,000 ADA	D1 addr1q9xm2dshrfkavnglluqhkcjcyfu32e89z4sctv86em8s5r4ma7gy2q2p5smffztajz6ry7azem76y34w2x4xkfkwcwzsvmvmgmm	C (unconfirmed)
00:54:44	same tx	550,000 ADA	D2 addr1qxtppdd8crtg92gsh6x3nhjpv2l8uf8smczkh45acqu2cwe26zut8ftflsuyyz5ecyvzxuakkq5z4yntn86qu0pr8xerq95wv7c	A, KuCoin
00:56:03	aac7c234...8b1c35577	550,000 ADA	D3 addr1qxt7t23qp0wpmp4q0xl89trxl8qpxhtkmhuy8ytc30qdjlscfa0dh5jhafsvsfp6cfn0cl8jmc56rjkfqpfr0stq9qv5ka	B (unconfirmed)
00:56:03	same tx	550,000 ADA	D4 addr1vyjw7r06ujhxckewckzuc7epgm5vhmgtctvnttcv7h4megzfgfj3 (enterprise)	D, Gate.io
01:10:12	38a6a022...00f9c5bf	236,851.540532 ADA	D5 addr1qye72ze9hd9p8yynejf4uwmzkm8hhrdkyttuc2h0ffzym3e7j3e3n8x8p0r3vgrl8cvv9wupvazm0z2kakc7jkghej9qsaj07n	A, KuCoin
02:00:06	fb1ce9fd...c02cd525	115,325 ADA (FLDT proceeds)	D6 addr1qy0pskzg5n42lgt5dhscnyz47g663f83x83j3tr0qj5042fr7gtja6rl0kldat0edxm2jdc9e80l2er220qvcjp2qewwuqe	B (unconfirmed)

Total to custodial clusters: 2,552,176.540532 ADA (KuCoin 786,851.54, Gate.io 550,000, clusters B and C 1,215,325). All six destination addresses are now empty.

6.3 OADA exit (1,988,322.18 OADA → FLDT → ADA on Minswap V2)

Time	Tx	What
01:52:30	ec1c6652...956d631	Minswap "Aggregator Market Order": 1,988,322.184396 OADA → FLDT (pool LP f5808c2c...ae7ba2c8...)
01:52:43	3ff0e818...9b4c06	Order executed: attacker receives 563,072.889266 FLDT (577f0b13...0014df10464c4454). OADA/FLDT pool goes from 132,317 OADA / 601,150 FLDT to 2,120,639 OADA / 38,077 FLDT
01:56:20	4ca68710...7ef8162	Market order: 563,072.889266 FLDT → ADA (pool LP f5808c2c...bc53f5c2...)
01:56:47	3e52dfce...babdb4	Executed: attacker receives 115,323.329399 ADA

The attacker realized only ~115k ADA for ~1.99M OADA; the OADA itself stayed in the Minswap OADA/FLDT pool (2,120,639 OADA immediately after the swap, 1,763,923 at 14:53 UTC after arbitrage wallets began pulling it out). What that means for a relaunch is in section 7.

6.4 Sweeps, wallet fingerprints and who controls the four clusters

We attach an exchange name to a wallet only when (a) it appears under that exchange's name in two independent public datasets: DefiLlama's exchange wallet lists (the CEX transparency data on [defillama.com](#), maintained in the public [DefiLlama-Adapters](#) repository) and the BEACNpool/ABCDE tracer-deposit dataset (participants deposited a tracer token through the exchange's own deposit flow and recorded where it landed), or (b) chain data shows the wallet paying directly into, or being funded directly by, such a wallet in the exchange's normal sweep pattern. Neither KuCoin nor Gate.io publishes its Cardano addresses on its own proof-of-reserves pages, so no label in this report is called exchange-published. An enterprise address (no stake key) is not evidence of an exchange by itself. Community tags are reported as tags, not as facts. No

public Cardano explorer (Cexplorer, Cardanoscan, Adastat) labels any wallet in this report, and none of the stake keys is registered or delegated.

Cluster	Destination → sweep tx (13 Sep, UTC)	Received	Fingerprint	Label and evidence
A	D2 → fb113bb4...5e6b29 01:10:12; D5 → 12711948...e6ae7a 01:20:33	786,851.54 ADA	1-in-1-out sweeps with a constant fee of 876,277 lovelace into consolidation wallet addr1q9h4f2vhh5...	KuCoin (two independent datasets plus a chain link). The consolidation wallet and the cold wallet it feeds are listed under KuCoin in DefiLlama's dataset (projects/kucoin/index.js , cardano.owners , entries added Jul–Aug 2025). The consolidation wallet is where an independent tracer deposit made to KuCoin was swept (ABCDE valid_deposits.csv , one hop). The withdrawal wallet that seeded the attacker (addr1qywgh46dqu7...) is funded by, and pays back into, that consolidation wallet; a copy of its address also sat in the DefiLlama KuCoin list from Jul to Oct 2025.
B	D3 → 881ca075...4fd6d 01:04:17; D6 → 13f6cde9...190ef01 02:09:53	665,325 ADA	fee 195,222; consolidation wallet addr1qjeag0c0q... , whose outbound pattern is tranches of 100k–800k ADA via passthrough addr1vy0j6cg02... into sink addr1vx7j284mq... ; both stolen tranches (549,999.80 and 115,324.80 ADA) are still unspent in it	Unconfirmed. No wallet in this cluster appears under any exchange in the datasets above. The sink addr1vx7j284mq... (76.86M ADA, one of the largest ADA addresses) is tagged as a major exchange's omnibus wallet by a community dataset (ABCDE finding F18, graded by its authors as inference, not proof), and two of the five addresses that project's participants used as deposit addresses for that exchange sweep their ADA into the same sink (our chain check, e.g. 615227ab... 13 Apr 2025 and 4a50f2c6... 28 May 2025). That exchange does not publish its Cardano addresses and has none in the DefiLlama dataset; we do not name it until it confirms.
C	D1 → 8e8ca8b3...a7a20b 01:04:57	550,000 ADA	fee 164,357; enterprise pass-through wallet addr1vydr3t70yt...	Unconfirmed, same sink as B. The pass-through wallet forwarded the 550,000 ADA into the same sink addr1vx7j284mq... at 01:07:10 (82da27f0...), as it does with everything it receives. Whether B and C are one exchange with two deposit systems or two venues sharing a custodian is not established.
D	D4 → 0474cf8c...73c653 01:00:12	550,000 ADA	enterprise destination address; 97-input batch consolidation into addr1qyywz67awv... , change to hot wallet addr1v8gxqvkpv...	Gate.io (two independent datasets plus a chain link). The consolidation wallet pays only into two addresses, the Byron DdzFFzCqrhtBatWqyFge... (until 5 Jun 2026) and addr1q9lxym523... (since 27 Dec 2025), both listed under Gate.io in DefiLlama's dataset (projects/gate-io/index.js , cardano.owners ; the Byron address since Nov 2022, addr1q9lxym523... since a Jul 2026 pull request submitted as an update of Gate.com's proof-of-reserves address list). The same cold wallet is where an independent tracer deposit made to Gate.io ended up (ABCDE).

Cluster-side wallets (state at 13 Sep 15:40 UTC; none of the stake keys is registered or delegated, which is typical of exchange custody):

Cluster	Role	Address	Balance	Active since	Label / source
A · KuCoin	consolidation (both stolen UTxOs passed through; spent at 02:00:42 and 07:04:15, section 7)	addr1q9h4f2vhh5vngqnsejan3psw6mj3a504fxlqm2eh3262qufesdvfs83uIr22vprsv9mwnt0vgkfwxflxkns32twqz dqjpq2na	773,053 ADA + thousands of token types	3 Mar 2023	DefiLlama KuCoin list; ABCDE tracer terminus (KuCoin)

Cluster	Role	Address	Balance	Active since	Label / source
A · KuCoin	withdrawal hot wallet	addr1qywgh46dqu7lq6mp5c6tzldpmzj6uwx335ydrpq8k7rru4q6yhkfqn5pc9f3z76e4cr64e5mf98aaeht6zwf8xl2nc9qr66sqg	716,776 ADA	3 Mar 2023	chain-linked: funded by and paying into the consolidation wallet above
A · KuCoin	cold wallet (receives from consolidation)	addr1x9qy3qeu59ngqfwe2uv4qz7n8y2wzzk0n5h3t0r2sdmflrzqfzpnegtqsqaj4ce2q9axwg5uy9vl8f0zk7x4qmk n7xq27ppet (script/multisig address)	8,295,915 ADA	11 Jan 2025	DefiLlama KuCoin list
B	consolidation	addr1q9jeag0c0qdpkux4u9y5um5w9z9lseqjwzgt9zhmtw2mfse7sc6vkfq p2302mycrqfglxsu2svde6m0yr03jgu uzgthqyytwpg	46,052,629 ADA (incl. the two stolen tranches, unmoved) + large NEWM, SNEK, HOSKY, MIN, BOOK, NIGHT, FLDT, INDY balances	3 Aug 2026	no public label
B	outbound passthrough	addr1vy0j6cg02l5u3wvyr4aefmadzk k2mcn5htpydcuwjtxjagszysey4	0 (forwards 100k–800k ADA tranches)	9 Jul 2025	no public label; forwards into the sink below
B + C	sink	addr1vx7j284mqe59w2mka36gf5xq0hvu8ms2989553fk5qh3prcapfpj3 (enterprise)	76,857,413 ADA	27 Aug 2024	community tag: exchange omnibus (ABCDE F18); receives sweeps from that exchange's tracer-deposit addresses; in no exchange list
C	passthrough	addr1vydr3t70yt8m8dqa4m46880p7v0kuv4m3h25d9qseykd8wcnhngOf (enterprise)	0 ADA	11 Feb 2026	no public label; forwards into the same sink
D · Gate.io	hot wallet	addr1v8gxqvkpy29nrcdn8rek3qvhj6alcldf2jfxlucam050fqv66ezq (enterprise)	476,009 ADA + NIGHT, HOSKY	5 Oct 2021	chain-linked: feeds the consolidation wallet below
D · Gate.io	consolidation	addr1qyywz67awvwrx38evlmjk5kj4k997ujtzsmjzuasyfw9r3zctzn5vv73j00yyu2r6re2t4j5z7lpf8y8mkyy528lrhs ugrz6t	550,000 ADA (the stolen tranche, unmoved)	25 Jun 2024	chain-linked: pays only into the two listed cold wallets below
D · Gate.io	cold wallets (receive from consolidation)	addr1q9lxym523pnns50dyymdv3n8vktyc0nkedr54c8d2prgxsumfyd8qtrg9yr4lqadjtczu6y94rtevjuua36rv75v0qsnstezggc ; Byron DdzFFzCqrhtBatWqyFge4w6M6VLgNUwRHiXTAg3xfQCudTcjJxSrPHVZJBsQprUEc5pRhgmWQaGciTsoZVwrSKmG1fneZ1AeCtLgs5Y	30,740,254 ADA; 0	24 Dec 2025; 23 Aug 2018 (last used 22 Aug 2026)	both in DefiLlama Gate.io list; ABCDE tracer terminus (Gate.io)

Label sources: DefiLlama-Adapters `projects/kucoin/index.js` and `projects/gate-io/index.js` (cardano.owners, checked 13 Sep 2026); BEACNpool/ABCDE `tracers/data/valid_deposits.csv` and `findings/F18_secondfi_incident.md`. DefiLlama's dataset (main at 0e722571, 13 Sep 2026) holds 93 Cardano addresses under 25 exchange names; none of the cluster B or C wallets is in it.

6.5 Prior activity on the same KuCoin account (9 Sep 2026)

Destination address D5 was not fresh: on 9 Sep 2026 it received six deposits totalling 1,257,453 ADA, each swept into the same KuCoin consolidation wallet with the same sweep fee. Exchanges normally assign deposit addresses per account, so the account that received 236,851 ADA of stolen funds is very likely the account that received 1.26M ADA four days earlier. Whether that earlier flow was the attacker's own funds or an unrelated

channel has not been established. The wallets behind it have been traced and will be taken up with the exchange; details are withheld while the trace is ongoing.

7. State at 13 Sep 2026, 16:20 UTC

Stolen ADA

At KuCoin, the UTxO from the D2 sweep was split at 02:00:42 ([af44b693...](#)): 502,672.87 ADA went to the KuCoin cold wallet as two UTxOs, still unspent, and 47,325.38 ADA into the withdrawal hot wallet. The UTxO from the D5 sweep was consumed at 07:04:15 ([e28dc539...](#)) together with hot-wallet float: 138,240.24 ADA went back to the withdrawal wallet and 302,571.20 ADA left KuCoin in a withdrawal to a third-party wallet. Exchange hot wallets pool customer funds, so chain data alone cannot say whose withdrawal that was; it is a question for KuCoin. Gate.io's consolidation wallet still holds the 550,000 ADA tranche as a single unmoved UTxO. Cluster C's 550,000 ADA reached the unconfirmed sink `addr1vx7j284mq...` (section 6.4) at 01:07:10. Cluster B's 665,325 ADA has not moved; it sits in the consolidation wallet `addr1q9jeag0c0q...` as two UTxOs. The attacker's own address holds 1.54 ADA.

Stolen OADA

The 1,988,322 OADA the attacker sold is in the Minswap V2 OADA/FLDT pool ([abd97295...#2](#), 14:53 UTC), which holds 1,763,923 OADA against 45,751 FLDT. The pool is thin and its implied OADA price is a small fraction of the peg. OADA has no protocol-level redemption, and every other OADA venue is close to empty: the Splash pool has 10 ADA; two further Splash stable pools at [addr1w9wnm7vle7al9q4aw63aw63wxz7aytnpc4h3gcjy0yufxwc3mr3e5](#) have 2.9 and 69.7 ADA; SundaeSwap V3 OADA/USDM has 31 ADA and OADA/ADA 53 ADA; Minswap V2 and V1 OADA/ADA have 9.5 and 6.1 ADA; Splash OADA/ENCS has 10 ADA. One wallet bought 1,213,726 OADA from the FLDT pool for 10,000 ADA at 02:09, has since sold 1,009,348 OADA back into the same pool for 9,094 ADA and holds 200,000 OADA. Any new ADA/OADA liquidity would be arbitrated against this stock.

The pools

The drained pool UTxO ([47dae2da...#0](#)) holds 10 ADA and 1,441,777.88 OADA with a negative tradable ADA reserve. The two further pools on the same validator hold about 72 ADA and 1,351 OADA in total.

8. Next steps

We keep tracing every wallet related to the attacker until each path reaches an exchange deposit wallet, so that the exchange can be asked for account information, and we will use other techniques as well. A separate report will follow. KuCoin and Gate.io will be contacted with the transaction IDs in sections 6.2, 6.4 and 7, including the 07:04:15 withdrawal; the operator of the sink in section 6.4 will be identified and contacted. If the attacker returns the funds through a direct channel to our team, we will treat it as a white-hat resolution, with a considered reward and without further escalation.